

Traditional Architecture

The above diagram shows the traditional three-layer architecture commonly used in software systems.

Layers in Traditional Architecture

1. Presentation Layer
2. Business Layer
3. Data Layer

1. Presentation Layer

The Presentation Layer is the front-facing layer of the application.

It includes:

- User interfaces
- Web applications
- Mobile applications
- Custom client applications

This is the layer where users interact with the system.

It collects user inputs and displays responses.

In simple terms, it acts as the front face of the application.

2. Business Layer

The Business Layer contains the application logic.

This is where:

- Servers run
- Business rules are executed
- Validations are performed
- Core application processing happens

It receives requests from the presentation layer, processes them, and interacts with the data layer when required.

3. Data Layer

The Data Layer is responsible for storing and retrieving data.

It includes:

- Databases
- Data storage systems

The business layer sends queries to this layer, and it returns the requested data.

How Traditional Architecture Works

In a traditional system, the flow is:

Client → Server → Generate SQL Query → Database → Return Data → Send Response to Client

Example:

If a client sends a request:

"Retrieve all orders for today"

Then:

1. The server generates an SQL query.
 2. The query is sent to the database.
 3. The database returns the result in JSON format.
 4. The server sends the formatted response (JSON or YAML) back to the client.
- All layers work together to complete the transaction.

New AI Architecture

In modern systems, we introduce an additional layer called:
AI Layer

This layer typically contains:

- Large Language Models (LLMs)
- AI services
- Intelligent processing components

Traditional Flow vs AI-Enhanced Flow

Traditional Flow

Client → Server → SQL Query → Database → Data → Client

AI-Enhanced Flow (Option 1)

Client → Server → SQL Query → Database → Data → LLM → Summarized/Processed Output → Client

In this case:

- The database returns raw data.
- The LLM processes, summarizes, or formats the data.
- The final refined output is sent to the client.

AI-Enhanced Flow (Option 2)

Client → Server → LLM (Generate Dynamic SQL Query)

→ Server → Database

→ Database returns data

→ Server → LLM (Generate required content or summary)

→ Server → Client

In this approach:

1. The LLM may generate the SQL query dynamically.
2. The database executes the query.
3. The returned data is again processed by the LLM.
4. The refined response is sent to the client.

Key Difference Between Traditional and AI Architecture

Traditional Architecture:

- Fixed logic
- Predefined queries
- Direct data retrieval and response

AI Architecture:

- Intelligent query generation

- Dynamic summarization
- Context-aware responses
- Natural language interaction

The AI layer enhances the system by:

- Reducing manual query writing
- Generating dynamic responses
- Improving user interaction
- Making the system more intelligent and flexible

Summary

Traditional architecture has three layers:

- Presentation
- Business
- Data

AI architecture introduces a fourth layer:

- AI Layer (LLM or intelligent system)

The AI layer can:

- Generate queries
- Process results
- Summarize information
- Improve decision-making

This evolution makes modern systems more intelligent, dynamic, and user-friendly.